



THE UNIVERSITY OF AUCKLAND
NEW ZEALAND

DATA PRIVACY PROTECTION

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COMPSCI 316



Data Privacy

Definition

I can decide:

when I'm disclosing my personal data

to whom I'm disclosing my personal data

how much personal data I'm disclosing



Data Privacy

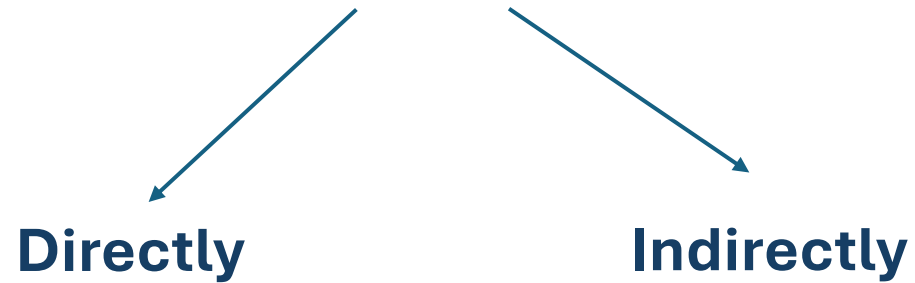
Personal Data

Any information which are related to an identified or identifiable natural person

Data Privacy

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Any information which are **related** to an identified or identifiable natural person



Data Privacy

Personal Data

Any information which are related to an identified or identifiable natural person

What is the **Problem**?

Data Privacy

Personal Data

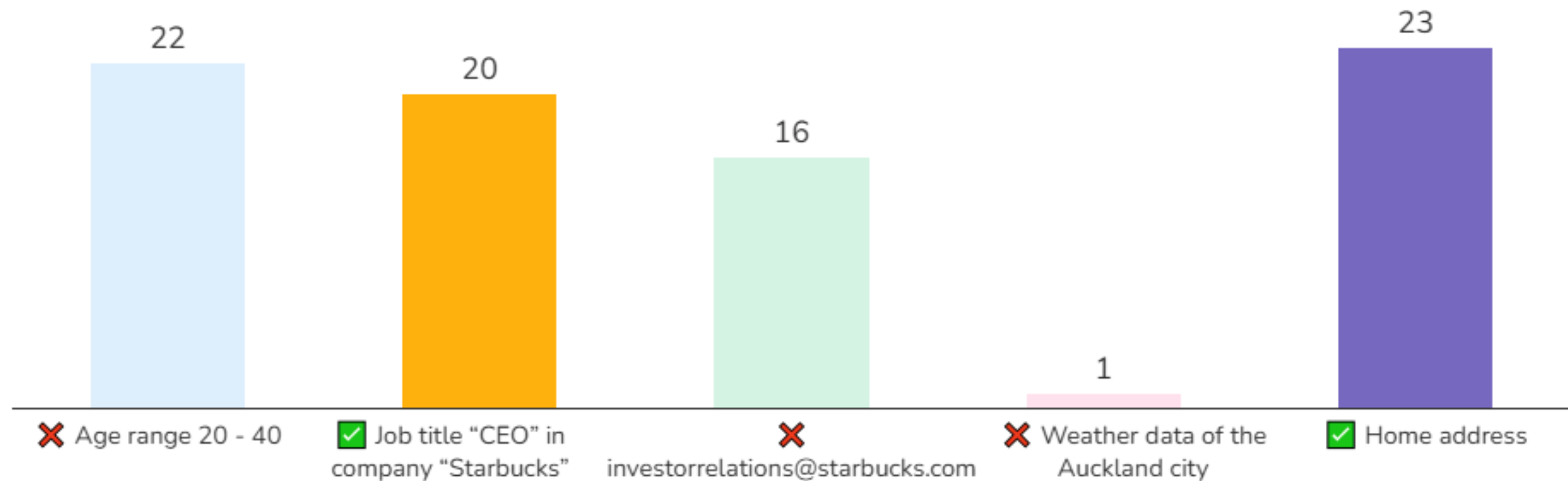
Any information which are related to an identified or identifiable natural person

What is the **Problem**?  Vague

Indirectly telling you to:

Be cautious when you are handling any type of information attached to a living person

Select the Personal Data



Data Privacy Protection

Amazing Mind Reader



[Amazing mind reader reveals his 'gift' - Duval Guillaume](#)

Exercise

Canvas - Be Mindful.....

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Breaktime for PowerPoint by Flow Simulation Ltd.

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Data Privacy Protection

Your Digital Footprint

The aggregated mass of data about you that is available online

1. What data do you **intentionally** share?
2. What are **hidden data** attached to your intentional data sharing activities (meta data)?
3. What records are collected regarding your **online activities**?
4. How can **other people** contribute to your digital footprint?
5. What **inferences** can be made from putting your data together?



Data Privacy Protection

Protection Mechanisms : End-User Perspective

- It is important to be mindful about your data sharing practices
- Researchers, educators, organisations try to improve awareness, knowledge, behaviour of users

Data Privacy Protection

Protection Mechanisms : End-User Perspective

What can you do?

- Read the privacy policies
- Use the privacy settings
- Ask to delete your data permanently if you no longer using a service
- Think about what data to share
- What technologies can you use?

VPN

Anonymous web browsers and search engines

Cookie blockers

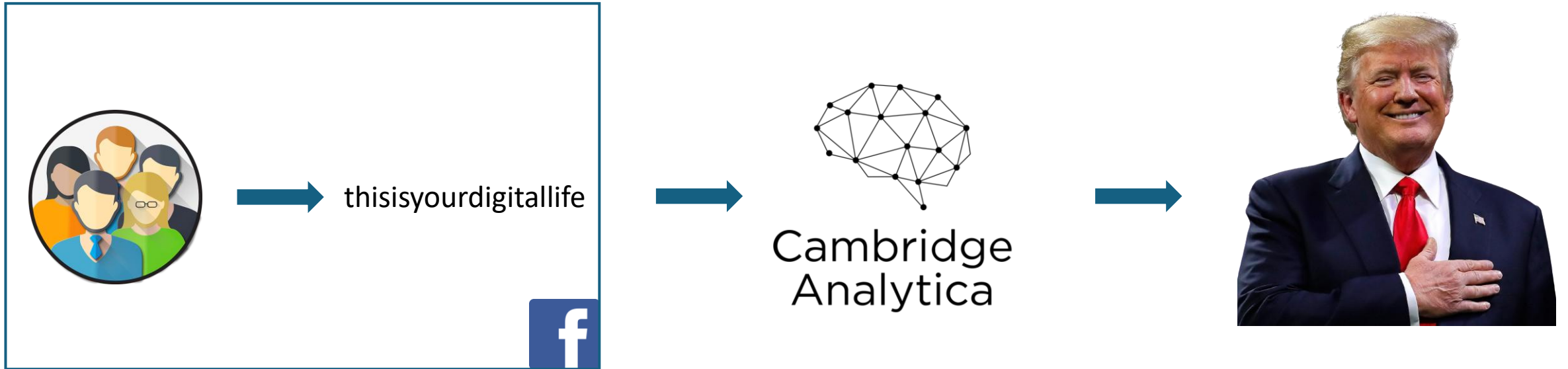
Incognito mode

Ad blockers

Is it enough?

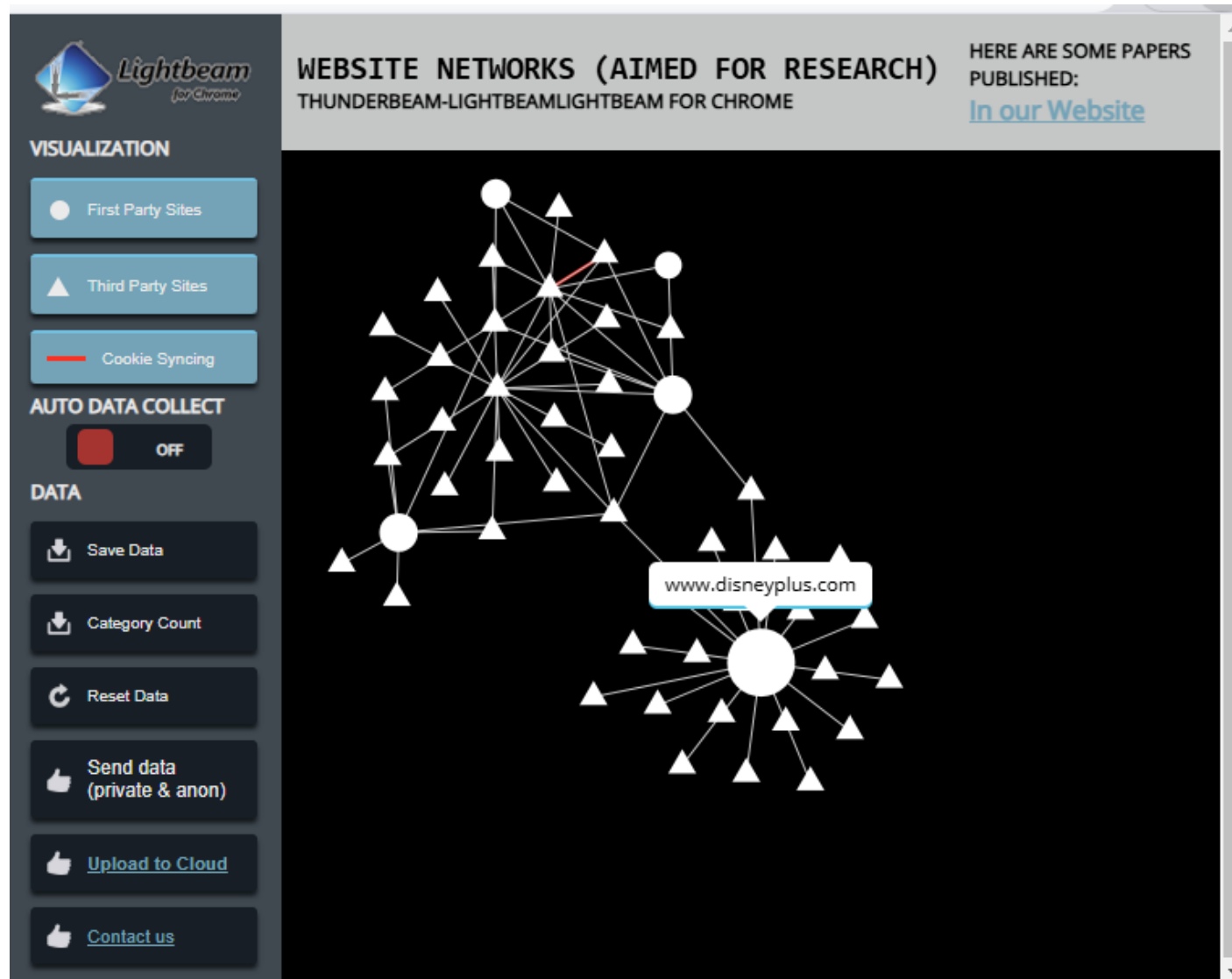
Case Study

Facebook Cambridge Analytica



Behind the Curtains

How your data is handled by the third-parties



Thunderbeam-Lightbeam for Chrome

Data Privacy Protection

End-User Perspective : Is It Enough?

Can you know what really happens to your data once they are disclosed?

- Services might not use robust data protection mechanisms
- Is security simply enough to protect your privacy in the digital world?
- Dark patterns in software

Stop sharing data?

Data Privacy Protection

Data Collection is Important

- Improve services and products
- Research and innovation
- Forensic analysis
- Rescue missions

Data Privacy Protection

Organisational Perspective

What happens if privacy protection is not considered from the offset of product development?

Can you envision how a business grow over time ?

- Amount of data collected
- Types of data collected
- Dependencies

Is it easy to do changes to an already developed product?

Data Privacy Protection

Organisational Perspective : Privacy by Design

Proposed by Ann Cavoukian

1. Proactive not reactive, preventative not remedial : **Plan it from the start. Don't wait until a privacy breach to happen.**
What are the advantages of this (other than better privacy)??
2. Privacy as the default : **By default, the products/services should be privacy protective**
3. Privacy embedded into design : **Priorities privacy in the product design. Privacy should be a core element in produce/service design**
4. Full functionality — positive-sum, not zero-sum : **Should not be a trade off for the user. Don't ever trade/exchange privacy for services.**

Example of breaking full functionality:

“Buy our premium” → “You won't see **targeted** ads”

Data Privacy Protection

Organisational Perspective : Privacy by Design

5. End-to-end security — lifecycle protection : **Should be applied to collection, process, share , store...**
as long as the data exists => during the entire data lifecycle
6. Visibility and transparency : **Communicate about the data handling practices openly to the product/service users**
7. Respect for user privacy — keep it user-centric : **Respect the user's right to privacy, be empathetic.**

Think :

“What will happen to my users if I use their data in this way?”,

“What will my users feel if I use their data in this way?”

Data Privacy Protection

Organisational Perspective : Privacy Threat Modelling

- A systematic way to identify potential threats to personal data
- Helps to understand how vulnerabilities can be exploited
- Before identifying the threats
 - Identify personal data handled by the product or service
 - Understand the flow of the identified data
- After identifying the threats
 - Impact assessment
 - Resolve the threats

Data Privacy Protection

Organisational Perspective : LINDDUN Threat Modelling Framework

Linking : Data items and user actions can be linked to **learn more** about the person even if it does identify a person

Identifying : **Identity** is revealed through leaks, deduction, inference

Non-repudiation : Someone can **attribute a claim to a person**.

<https://linddun.org/threats/>

Detecting : Arrive at a **conclusion** about a person through **observation**



Data disclosure : **Excessively** collecting, sharing, processing, storing personal data

Unawareness & Unintervenability : Insufficient transparency, feedback to the users or less involvement of the **user**

Non-compliance : Violation of best practices, standards and regulations

Exercise

Canvas – Track the Privacy Threats

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Data Privacy Protection

Organisational Perspective : Challenges

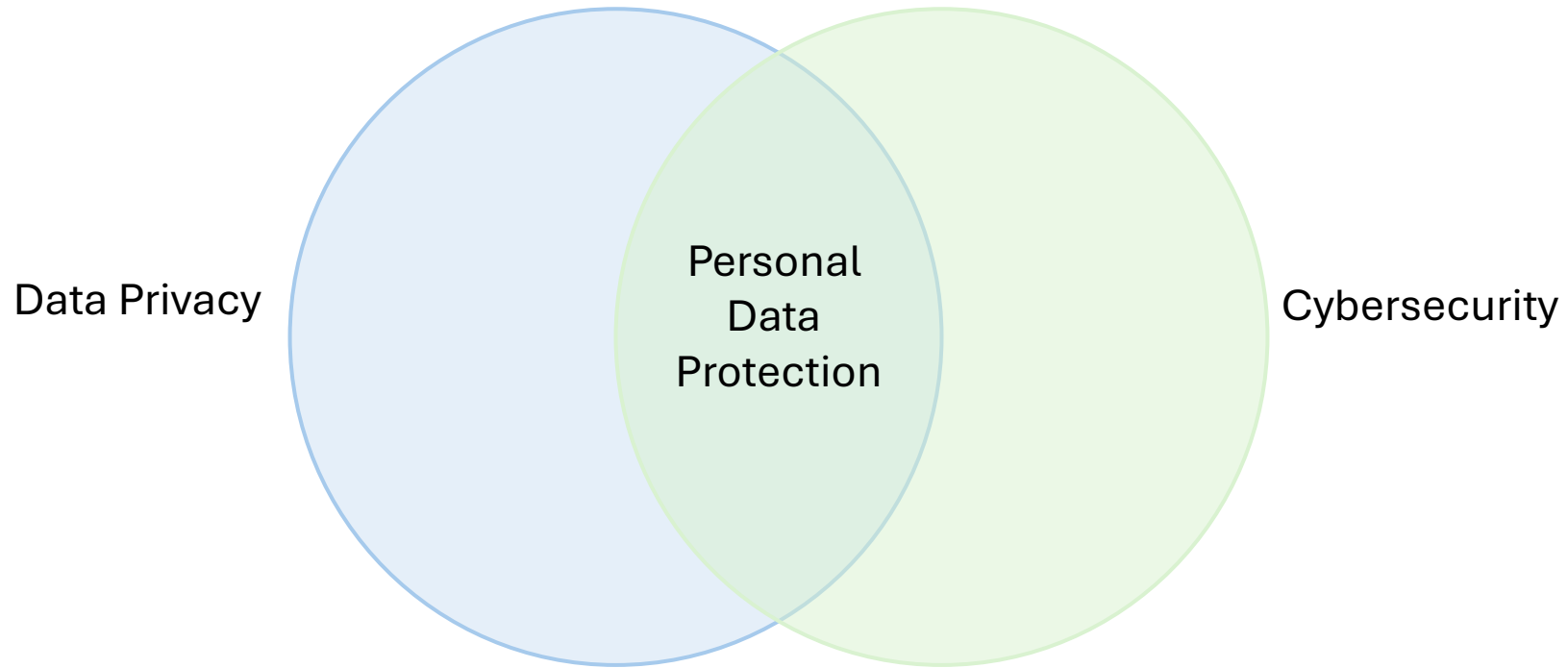
- Lack of support from the higher ups
- Lack of technical guidelines for privacy protection
- Lack of resources – money, time, manpower, expert support
- Lack of awareness and knowledge of the employees
- Lack of privacy mindset of the employees

Data Privacy Protection

Organisational Perspective : Solutions

- Training : Gives employees knowledge and skills about privacy
- Recruit privacy experts – ex: lawyers, chief privacy officers, researchers, developers with good privacy knowledge (privacy champions)
- Give incentives to privacy champions : promotions, salary increments (improves the motivations to protect privacy)
- Enforce policies within the organisation : Govern the privacy practices of the employees
- Standardise privacy : Employees will realise privacy is an important part in product development
ex: Making privacy requirement gathering compulsory

Data Privacy & Cybersecurity



Summary

- Understanding personal data
- User perspective of privacy protection
- Organisational perspective of privacy protection
- Organisational challenges in answering privacy consideration



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